A 336Gb/s/lane 2.47pJ/b Integrated Transmitter with Silicon Photonic TW-MZM and BW-Boost High-Linearity Driver

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Abstract: A hybrid-integrated optical transmitter comprising of a SiPh TW-MZM wire-bond with co-designed BW-boosted driver in SiGe BiCMOS is presented. Experimental results demonstrate 336Gb/s/lane PAM-8 optical eye diagrams at 2.47pJ/bit power consumption.

1. Introduction

The rapid development of AI is driving the demand for optical interconnects with ever-growing data rate. Silicon photonics (SiPh) has been proposed as an attractive solution for next-generation energy-efficient, high-bandwidth and low-cost interconnects in datacenters. Currently the channel speed is advancing from 224Gb/s towards 400Gb/s per lane. Recent publications have demonstrated 320Gb/s optical transmitter using TFLN modulator [1] or 336Gb/s MZM alone in silicon [2]. Instead of further increasing the baud-rate, multi-level Pulse-Amplitude Modulation (PAM-6, PAM-8) can further scale the data-rate without additional wavelengths or higher BW. However, it requires wideband driver circuits with higher linearity.

We present a hybrid-integrated SiPh MZM transmitter, consisting of a 90nm SiGe BiCMOS driver wire-bonded to a 40nm SiPh MZM, which incorporates the following innovations: (1) a 2.5mm-long TW-MZM fabricated on a 300mm SiPh platform, achieving 41 GHz electro-optic (EO) BW under 3V low-voltage bias, (2) a distributed continuous-time linear equalization (CTLE) driver co-designed to extend the MZM E/O BW, while delivering $3.2V_{ppd}$ swing and maintaining high linearity. (3) E/O integrated multi-channel transmitter with high signal integrity at 336Gb/s/lane high-speed.

2. Proposed Hybrid Integrated Transmitter

Figure 1 illustrates the block diagram and micro-photo of the proposed SiPh transmitter. It consists of a quad-lane arrayed driver in SiGe BiCMOS wire-bonded with a TW-MZM fabricated in 300mm SiPh process. The TW-MZM exhibits a high modulation efficiency of 1.5V·cm but relatively limited 41GHz BW. To overcome this BW limitation, the co-designed driver integrates a 4.1dB tunable CTLE that peaking at 62GHz, extending the overall E/O BW to support 112GBaud operation. The driver's back termination impedance (40Ω) is designed to be slightly higher than the MZM (33Ω) to save power, while maintaining a reflection coefficient below -20dB. As shown in Fig. 1, the driver's common-mode output voltage is biased by both the driver and the MZM sides and stabilized at 3V through an internal common-mode feedback loop to ensure maximum output swing. This scheme reduces power consumption by approximately 25% compared to driver-side-only biasing. In addition, the transmitter supports scalable multi-channel for high-density data transmission.

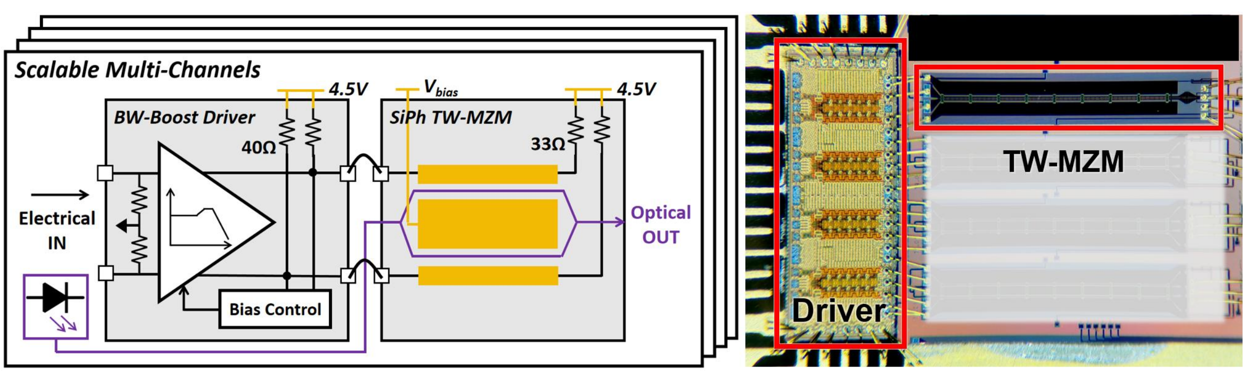

Figure 1. Block diagram and micro-photo of the proposed SiPh transmitter.

3. High linearity TW-MZM on 300mm SiPh platform

The SiPh TW-MZM was fabricated on a 300mm SOI wafer with a 220nm thick silicon layer and a 2µm thick buried oxide (BOX). The mode profile of the TE_0 in the ridge waveguide and the cross-section of TW-MZM are shown in Fig. 2(a). For velocity matching between electrical and optical signals, the modulator employs a ground-signal-signal-ground (GSSG) electrode configuration that integrates a capacitive loaded traveling-wave (CLTW) structure (Figure 2(b)). The MZM characteristic impedance is designed for 65Ω, enabling the co-packaged transmitter to achieve improved signal integrity. The TW-MZM consists of two 2.5mm-long phase shifters, each equipped with an embedded p-n junction. These two p-n junctions are serially connected to form a push-pull configuration. Customized doping concentrations enable the MZM to achieve high linearity, supporting high-order signal modulation formats. Photographs of a 300mm SOI wafer and the fabricated TW-MZM are displayed in Fig. 2(c). TW-MZM delivers median EO bandwidths of 40.8GHz (≈ 41GHz) and 52.9GHz (≈ 53GHz) under bias voltages of −3V and −6V, respectively. Fig. 2(d) presents the EO S_{21} and S_{11} curves, where the numerical simulation results are consistent with the experimental test results.

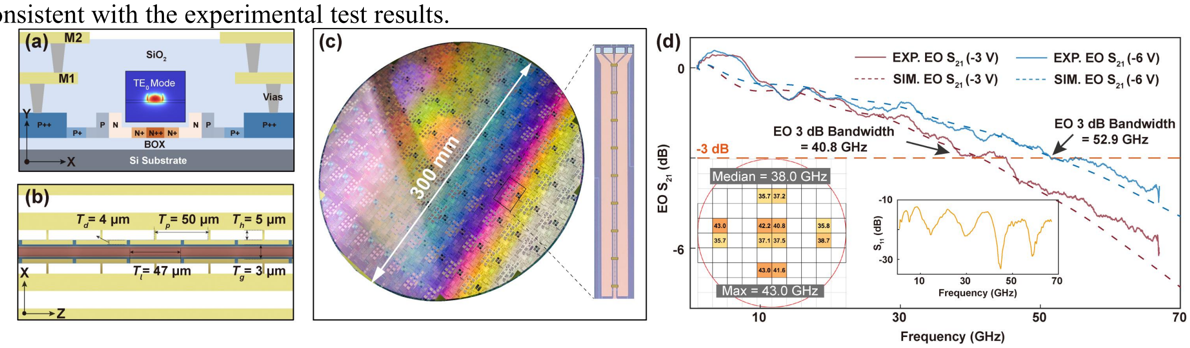

Fig. 2. (a) Simulated mode profile of the ridge waveguide and the cross-section of CLTW-MZM; (b) Schematic view of the periodic CLTW electrode structure. (c) Optical photos of fabricated 300 mm SOI wafer and CLTW-MZM. (d) Electro-optic response of the CLTW-MZM.

4. The BW-boost high-linearity Driver

The MZM driver incorporates two key design features to enable 336Gb/s PAM-8 E/O modulation. First, a distributed topology is adopted to extend BW. The distributed driving stage comprises six gain cells, while artificial transmission lines (ATLs) absorb the input and output capacitances of gain cells [3]. Each gain cell integrates series-peaking inductors for further BW extension. A 62GHz, tunable 4.1dB CTLE is embedded in the driver using a source degeneration resistor and a tunable shunt capacitor (Re and Ce in Fig. 3(a)). As shown in Fig. 3(e), the CTLE compensates for the high-frequency loss of the MZM, extending the overall E/O BW from 41GHz to above 70GHz. Second, a modulated-base structure (Fig. 3(a)) enhances the linearity of the gain cells by distributing the output swing across transistors Q1 and Q2, unlike a conventional cascode topology where Q2 bears the full swing [4]. The 12Ω resistor Re further improves linearity through local feedback. Figure 3(d) shows a simulated total harmonic distortion (THD) of 1.5% at a 3.2Vppd output swing.

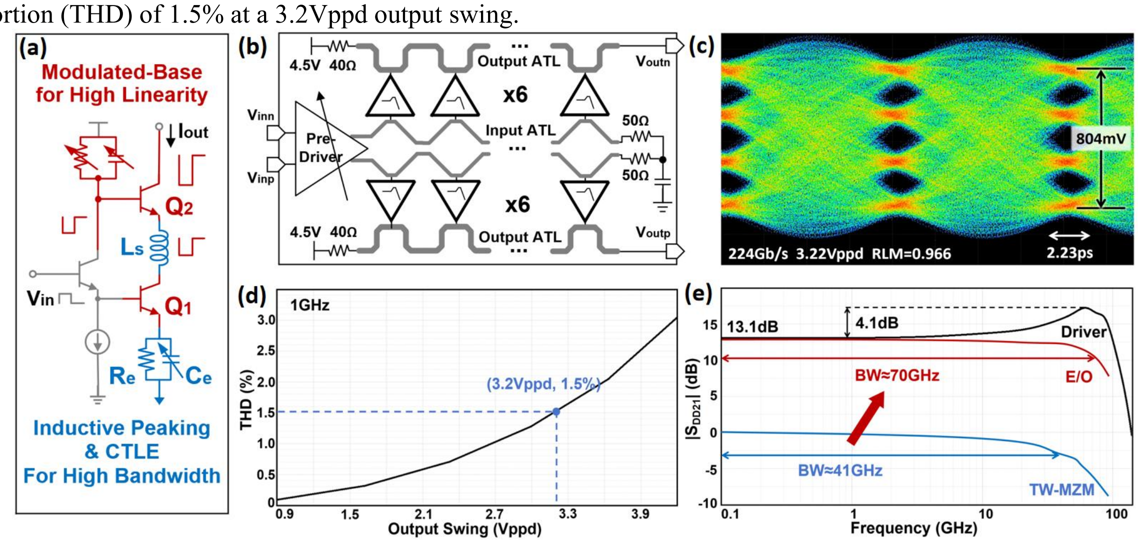

Figure 3. (a) One leg of the gain cell; (b) Simplified block diagram of the driver; (c) Measured 112GBaud PAM-4 electrical eye patterns (with 12dB attenuator); (d) Simulated THD at 1GHz; (e) Simulated small-signal forward gain and peaking at 62GHz.

5. Experiment Results

Time-domain test is performed to characterize the MZM driver. The arbitrary waveform generator (AWG, Keysight M8199A) provides a PRBS13 serial input pattern, which is applied to the MZM driver via on-chip probing. The amplified output is then extracted via probes and captured by a high-speed oscilloscope (Keysight N1046A) with 10-tap feed-forward equalization (FFE) enabled. A 12dB attenuator is inserted in the signal path to prevent the oscilloscope from overloading due to the large output swing. As shown in Fig. 3(c), at 224Gb/s PAM4, an output swing of 3.22Vppd is achieved with a ratio level mismatch (RLM) of 96.6%.

The high-speed E/O measurement setup is shown in Fig. 4(a). The input signal configuration remained consistent with electrical test. The driver's output is wire-bonded to the TW-MZM with matched footprints to deliver a 3.2Vppd linear output swing. The output optical is captured using an optical oscilloscope (Keysight N1032A) with 10-tap FFE enabled. Figure 4(b) shows a measured NRZ optical eye pattern at 112Gb/s with an extinction ratio (ER) of 4.05dB. Figure 4(c) presents a PAM4 eye at 224 Gb/s with an ER of 3.38 dB and an RLM of 94.4%. Figure 4(d) and 4(e) display 112GBaud PAM6 and PAM8 optical eyes, respectively, achieving a maximum data rate of 336Gb/s. The total power consumption of the transmitter in this test is 831mW.

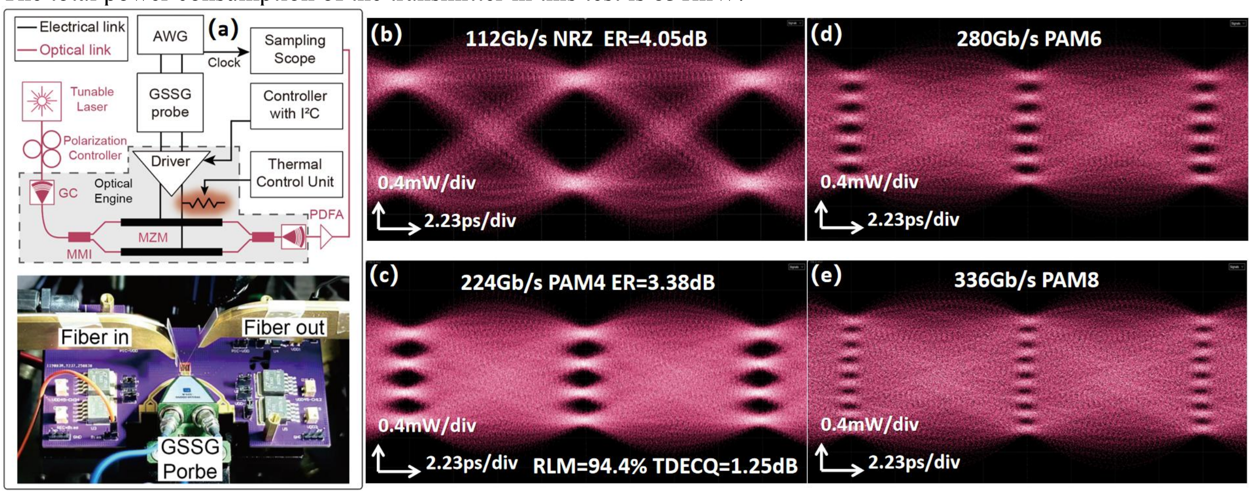

Figure 4. (a) Test setup; Measured (b) 112GBaud NRZ (c) 112GBaud PAM-4 (d) 112GBaud PAM-6 (e) 112GBaud PAM-8 optical eye patterns.

6. Conclusion

This paper demonstrates a transmitter integrating a 90nm SiGe BiCMOS electrical driver with a TW-MZM fabricated on a 300mm SiPh platform. The integrated CTLE in the driver effectively extends the overall E/O bandwidth, and the combined circuit and device innovations enable the high system-level linearity essential for PAM-8 operation. Measured results show that the transmitter achieves a maximum data rate of 336Gb/s with a total power dissipation of 831mW, corresponding to an energy efficiency of 2.47pJ/b.

7. Acknowledge

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8. Reference

[1] Declercq, Jakob, et al. "320 Gb/s Unamplified Transmission Using 100 GHz Ge PD and TFLN MZM on a Foundry-Compatible SiPh Platform Co-Packaged with Traveling-Wave Drivers and TIAs," The 51st European Conference on Optical Communications (ECOC), 2025.
[2] E. Jia et al., "A 336 Gbps Traveling-Wave Mach-Zehnder Modulator in a 300 mm Silicon Photonic Platform," 2025 Optical Fiber Communications Conference and Exhibition (OFC), 2025, pp. 1-3.
[3] M. H. Mahmud, H. Al-Rubaye and G. M. Rebeiz, "Broadband Linear Drivers for 800G/1.6T Energy Efficient Optical Links," 2024 IEEE BiCMOS and Compound Semiconductor Integrated Circuits and Technology Symposium (BCICTS), 2024.
[4] K. De Bruyn et al., "Linearity Enhancement Analysis of Breakdown Voltage Doubler and Demonstration in 60 GBd SiGe BiCMOS Driver," 2024 IEEE International Symposium on Circuits and Systems (ISCAS), 2024.
[5] IEEE Standard for Ethernet, IEEE Std 802.3-2022, clause 94. IEEE, 2022.